

Muhammad Daffa Adika Utama

Computer Engineering student at Universitas Pendidikan Indonesia with a concentration in Intelligent Device Development. Proven track record in developing AI-powered systems and IoT solutions, evidenced by multiple relevant projects. Skilled in AI Engineering and Full-stack Development, with a focus on creating impactful technology for social and environmental challenges.

Kota Bekasi, Jawa Barat | daffadikau@gmail.com | +6281386414411 | [linkedin.com/in/Daffadikau/](https://www.linkedin.com/in/Daffadikau/) | <https://github.com/Daffadikau/>

EDUCATION BACKGROUND

Universitas Pendidikan Indonesia

Undergraduate Bachelor of Computer Engineering. GPA 3.65 out of 4.00

Bandung, Jawa Barat
Aug 2023 – 2027 (Expected)

Specialized in Intelligent Device Development with hands-on experience in real-time AI deployment, high-dimensional data modeling, and hardware-software integration through various national-level research projects.

SMAI PB. Soedirman

High School Diploma in Information Science Major

Bekasi, Jawa Barat
July 2020– July 2023

Graduated from the Science Major with a specialized focus on Science and Technology. Developed strong analytical thinking and problem-solving skills through a rigorous curriculum.

PROFESSIONAL EXPERIENCES

PT IROSTECH Solusi Intelejen

Backend Developer and AI Engineer Intern

Bandung, Jawa Barat
Feb 2026 – Aug 2026

- Developing a CBRN (Chemical, Biological, Radiological, and Nuclear) Monitoring Platform to facilitate real-time surveillance and data visualization from unmanned aerial vehicles (UAVs/Drones).
- Engineering robust Backend Architectures to handle high-frequency data streams and sensor telemetry, ensuring low-latency communication between drones and the command center.
- Implementing AI-driven Analytics to automate the detection and assessment of hazardous materials, enhancing the decision-making process for emergency response.

ORGANIZATIONAL & VOLUNTEER EXPERIENCES

Rumah Prestasi Kemahasiswaan Universitas Pendidikan Indonesia

Staff of Research and Creativity

Bandung, Jawa Barat
Jan 2026 – Currently

- Curated and mentored high-impact digital innovation projects for the National Student Digital Innovation Competition (LIDM), focusing on technical feasibility and research integrity.
- Conducted rigorous internal reviews of research methodologies and system architectures to ensure alignment with national competition standards.
- Facilitated brainstorming sessions and technical workshops to enhance the creative output and problem-solving capabilities of multidisciplinary student teams.

Himpunan Mahasiswa Teknik Komputer Universitas Pendidikan Indonesia

Head of Staff Pengabdian

Bandung, Jawa Barat
Jan 2025 – Jan 2026

- Orchestrated BASKOM (Bakti Sosial with Tekkom), a flagship social welfare program, successfully mobilized and distributed over IDR 5,000,000 in humanitarian aid. Managed end-to-end fundraising campaigns and donation distributions for local orphanages, natural disaster victims, and university peers in financial need.
- Managed targeted fundraising campaigns for diverse social causes, including coordinating emergency funds for the Bekasi Floods and Sumatra Natural Disasters, organizing Ramadhan charity events while providing financial support for local orphanages, and facilitating internal solidarity funds for students in financial distress.
- Directed the end-to-end logistics of donation cycles, ensuring financial accountability and transparent reporting to all stakeholders and donors.

Korps Sukarela Palang Merah Indonesia Universitas Pendidikan Indonesia

Staff of Bakti Sosial

Bandung, Jawa Barat
Sep 2024 – Jan 2026

- Acted as a First Responder and Field Medic during emergency missions and natural disaster relief operations, providing immediate medical assistance and logistical support in high-pressure environments.
- Spearheaded the Blood Donation Program in collaboration with the Bandung Regency Red Cross (PMI Kabupaten Bandung), successfully managing large-scale donor recruitment and event logistics.

- Demonstrated exceptional adaptability and physical stamina while performing field duties in challenging geographical and environmental conditions
- Cultivated and maintained strategic partnerships with regional Red Cross chapters to enhance coordinated disaster response and community health initiatives.

PROJECT

Optical Mark Recognition Mobile Application – Group Project

Universitas Pendidikan Indonesia

Subject Intelligent Device Systems, 4.00/4.00

Sep 2025 – Jan 2026

- Engineered an automated grading system utilizing advanced pattern recognition and image preprocessing techniques. Reduced manual grading time by 90% while maintaining a 99,7% accuracy rate
- Designed algorithms for feature localization and document alignment to precisely classify human-marked data from scanned academic documents.

Automated IOT Based Drying Shoes Closet – Group Project

Universitas Pendidikan Indonesia

Subject Consulting Project, 3.75/4.00

Sep 2025 – Jan 2026

- Engineered a smart drying cabinet for a local shoe laundry business to optimize the drying process and prevent material damage from overexposure to heat.
- Integrated temperature and humidity sensors with a microcontroller-based control system to automate airflow and heating elements.

Community Service Trias Bakti Program – Work Program

Desa Drawati

Ideas Division on Backend Developer and Systems Architect

Feb 2025 – Jul 2025

- Spearheaded the digital transformation of local Micro, Small, and Medium Enterprises (MSMEs) in Drawati Village by developing a dedicated web-based platform to expand market reach.
- Designed and documented a comprehensive Software Requirement Specification (SRS) to ensure the system architecture met the specific logistical and operational needs of rural businesses.
- Engineered robust Backend Services and database schemas to manage product inventories and transaction records efficiently.
- Facilitated technical onboarding for local stakeholders, bridging the gap between complex software systems and user-friendly community applications.

SKILLS & LANGUAGES

Skills: Python, Typescript, JavaScript, Dart, HTML, CSS, PHP, SQL, Firebase, AWS, Azure, Machine Learning, Data Analysis, Full-stack Development, Mobile Development, Backend Development, Data Structure and Algorithms, Internet of Things (IoT), Microcontroller.

Languages: Indonesian (Native), English (Full Professional Proficiency)

ACHIEVEMENTS AND AWARDS

- 2nd Place Winner Best UI/UX Front-End Design at Himpunan Mahasiswa Universitas Pendidikan Indonesia, 2025
- Best Department of the Year (P2M) at Himpunan Mahasiswa Universitas Pendidikan Indonesia, 2026
- Best Biro of the Year (Pengabdian) at KSR PMI Unit UPI Kampus Cibiru, 2026
- Staff of the Month (July) at KSR PMI Unit UPI Kampus Cibiru, 2025
- Biro of the Month (September) at KSR PMI Unit UPI Kampus Cibiru, 2025
- Best Robotics and IT Student Program at SMAI PB Soedirman Bekasi, 2022